A pre-registered field test of Seismic Frequency Resonance (SFRT) targeting for orogenic gold: dual-impedance re-analysis, georeferenced structural integration, and drill-test protocol for the Lac Courville Property, Abitibi, Québec

Aimin Xue[1] and Ruqing Chen[2]

[1]Beijing Petrosound Geoservices Corp., Beijing, China. aiminx@126.com
[2]Minesound Ltd., Montreal, Canada. ruqing@hotmail.com

Pre-registration manuscript — Part I of a two-part test (predictions filed before drilling)

Abstract

Context. Claims that a geophysical method "predicts" mineralization are usually published *after* drilling, leaving them open to hindsight bias. We remove that ambiguity by registering, before any drilling, the targets, drill-hole geometry, and explicit pass/fail criteria for a single Seismic Frequency Resonance (SFRT) profile (Line 1) over the Lac Courville gold property, eastern Abitibi greenstone belt, Québec. **Methods.** We re-process co-located P- and S-wave impedance volumes (Zp, Zs) acquired along Line 1, compile and georeference the documented gold record of Courville Township from 31 historical assessment reports and a 2021 NI 43-101 report, and integrate the two. **Results.** The Zp/Zs ratio — a density-independent fluid/shear discriminant not used in the original survey — defines three sub-line targets, the strongest being T1 (DAS $\approx$11,500–12,000; 250–550 m). Georeferencing the GM-49463 compilation via its lat/long graticule lets the Manneville Fault be digitized to UTM (strike N61°W); it bounds the gold corridor $\sim$1.2 km south of Line 1, which therefore sits on the favourable hanging-wall flank where historical mapping places auriferous branch shears. The NW-trending (N45°W) gold-branch projects onto the northeast-central part of Line 1. **Test.** Three 600 m diamond-drill holes (azimuth 225°; dips $-58°$ to $-65°$) are designed to intersect T1–T3. We pre-define a structural-hit criterion (a shear/altered/veined zone within $\pm$50 m of the predicted depth window) and a mineralization-hit criterion (Au $\geq$0.3 g/t over $\geq$1 m). Independent QP core logging, ISO-accredited assaying, and RTK collar survey will adjudicate the outcome in Part II. **Significance.** The design provides a falsifiable, externally witnessed test of whether SFRT adds value for gold targeting, independent of the result.

Data and code availability: all data, processing code, figures and this manuscript are archived at the project repository (GitHub) with a versioned DOI (Zenodo); see §10. **Competing interests:** R. Chen is affiliated with the property holder (Minesound Ltd.); A. Xue is affiliated with the SFRT method provider. The pre-registered protocol and third-party verification (§8) are designed to mitigate this.

1. Introduction

Geophysical exploration routinely produces "targets" that are later drilled, and the literature is rich in case studies in which a method is reported to have successfully predicted mineralization. Because such reports are almost always written after the drill results are known, they cannot exclude post-hoc rationalisation: targets, tolerances, and success criteria can be adjusted, consciously or not, to match the outcome. The result is a persistent credibility gap for newer or proprietary methods, for which an independent, pre-committed test is rarely available.

This paper is the first of a two-part study designed to close that gap for one method — Seismic Frequency Resonance (SFRT) — on one property. Here (Part I) we fix, in advance of drilling, (i) the geological and geophysical basis for three targets, (ii) the drill-hole geometry that will test them, and (iii) explicit, quantitative criteria by which the test will be judged a success or a failure. Part II will report the drill results and adjudicate them against these criteria without modification. By time-stamping the predictions (§10) we make the test falsifiable in the strict sense: a defined class of drilling outcomes would count against the method.

We deliberately separate two questions that are often conflated. The first is whether SFRT images *structure* — the shear zones, alteration, and competency contrasts that localise orogenic gold. The second is whether a given imaged structure actually *carries gold* at a particular intersection, which additionally depends on the local fertility of that structure. SFRT, like all structurally-vectoring geophysics, can in principle succeed at the first while failing at the second in any single hole. We therefore register the two criteria separately and assess them separately.

2. Regional and district geology

The Lac Courville Property lies in the southeastern Abitibi Subprovince of the Superior Province, Canadian Shield, within ∼20 km of the Val-d'Or mining camp — one of the most productive orogenic-gold districts on Earth. The Abitibi greenstone belt formed in the late Archean (∼2.7 Ga), with the principal gold-forming events at ∼2.68–2.67 Ga, and its greenstone-hosted quartz-carbonate vein deposits are the type examples of the orogenic-gold class (Dubé and Gosselin, 2007).

The property straddles the northern contact of the *Pascalis-Tiblemont batholith*, a large multiphase, synvolcanic intrusive complex of diorite, granodiorite and tonalite, against mafic-to-felsic metavolcanic rocks of the Harricana Group (Globex Mining Enterprises Inc., n.d.; LaFleur Minerals Inc., 2026). This contact is occupied by the *Manneville Deformation Corridor*, a regional structure along the northern margin of the batholith that is associated with numerous gold occurrences and is interpreted as the eastward continuation of the Manneville Fault (Globex Mining Enterprises Inc., n.d.; LaFleur Minerals Inc., 2026; Bischoff, 1965). Stratigraphy and the principal structural grain trend WNW–ESE; the gold-controlling shears dip steeply to the northeast.

District gold mineralization is structurally controlled and intrusion-hosted: free gold in quartz (± calcite ± tourmaline) veins and veinlets within silicified, pyritized diorite and granodiorite, with local chalcopyrite (Globex Mining Enterprises Inc., n.d.; Laverdière, 2021). This style — and its setting on a second-order corridor splaying from a crustal-scale break — is characteristic of the Val-d'Or orogenic-gold environment (Dubé and Gosselin, 2007).

3. Property geology and exploration history

Gold has been documented along the Manneville/Pascalis-Tiblemont corridor immediately northwest of, and along strike toward, the property for over seventy years. The historical record (31 Québec assessment reports, 1947–2021; compiled in the project repository) establishes a consistent, structurally-controlled gold system. Key documented endowment includes: the Pershing-Manitou deposit (45,350 t at 7.9 g/t Au) (Beauregard, 1990); the Courville Gold Project "Zone A" stockwork (1–7 g/t Au; free gold to ∼9 g/t; five zones

$>$0.10 oz/t defined by 1980s drilling), centred $\sim$3.5 km west of the property (Patel, 1988; Laverdière, 2021); and the Uniacke/Obradovich drilling, including hole VC-10 (1.1 ft at 0.68 oz/t, $\approx$23.3 g/t Au) (Hawley, 1994). Historical mapping traces gold-bearing porphyry dykes and the Manneville Fault southeastward across the southern part of the property, south of Lac Courville (Bischoff, 1965).

Critically for the present test, the 2021 NI 43-101 report on the adjacent Courville Gold Project concludes that the deposit comprises en-échelon veins of limited individual extent and that the project's principal unresolved problem is an *undefined structural framework* — specifically, the absence of an identified major controlling fracture — and recommends a dedicated structural study (Laverdière, 2021). Imaging crustal-scale shear zones and their dilational intersections is precisely the information that report identifies as missing, and is the stated purpose of the SFRT survey analysed here.

4. SFRT method and Line 1 data

SFRT is a passive-source seismic technique that uses the frequency-resonance (transfer-function) response of transmitted waves through solid media to recover wave-impedance variation, a parameter reported to be more sensitive to lithology and structure than seismic velocity (Xue et al., 2020; Xue, 2023). We treat the method's physical basis as a hypothesis under test rather than as established, consistent with the falsifiable design of this study.

Line 1 yields co-located P- and S-wave impedance volumes (Zp, Zs), each 176 stations (DAS 10,000–13,500 at 20 m spacing) by 750 depth samples (0–1498 m at 2 m). Because $\mathrm{Zp} = \rho V_p$ and $\mathrm{Zs} = \rho V_s$, the ratio Zp/Zs equals V_p/V_s and is independent of density, making it a lithology/fluid/alteration discriminant. In the Line 1 data Zp/Zs clusters at 1.40–1.55 (median 1.43), well below physical rock V_p/V_s (1.7–1.9); the impedances are therefore *relative* (uncalibrated), and all interpretation uses depth-detrended relative anomalies rather than absolute values. The composite prospectivity combines relatively low Zp (alteration/fracture/fluid $\rightarrow$ low velocity), relatively high Zp/Zs (fluid-saturated shear damage), and high lateral gradient (structural contact / dilational site).

5. Results: sub-line targets

The four-panel re-analysis (Fig. 1) adds two products absent from the original single-impedance interpretation: a low-Zs (shearing/fracturing) zone at DAS $\approx$11,700–12,000, and a strong high-Zp/Zs (fluid/alteration) anomaly at DAS $\approx$11,500–12,000, 250–550 m, seated on the flank of a competent high-Zp body — a classic competency-contrast dilation site. Three targets are defined (Table 1), ranked T1 $>$ T2 $>$ T3. A western near-surface anomaly (DAS 10,000–10,460, 0–170 m) carries a low score and low Zp/Zs and is interpreted as an overburden/line-edge effect, not a target.

Table 1: SFRT targets on Line 1 (depth-detrended Zp/Zs anomaly).

Target	DAS	Depth (m)	UTM centre (18N)	Δ(Zp/Zs)	Interpretation
T1 (primary)	11,500–12,000	250–550	326,054 / 5,356,352	+0.065	competent-body flank + fluid
T2	12,380–12,840	340–660	325,538 / 5,355,632	+0.045	F3–F4 corridor
T3	13,240–13,340	280–500	325,135 / 5,355,067	+0.045	F4 east-flank fault

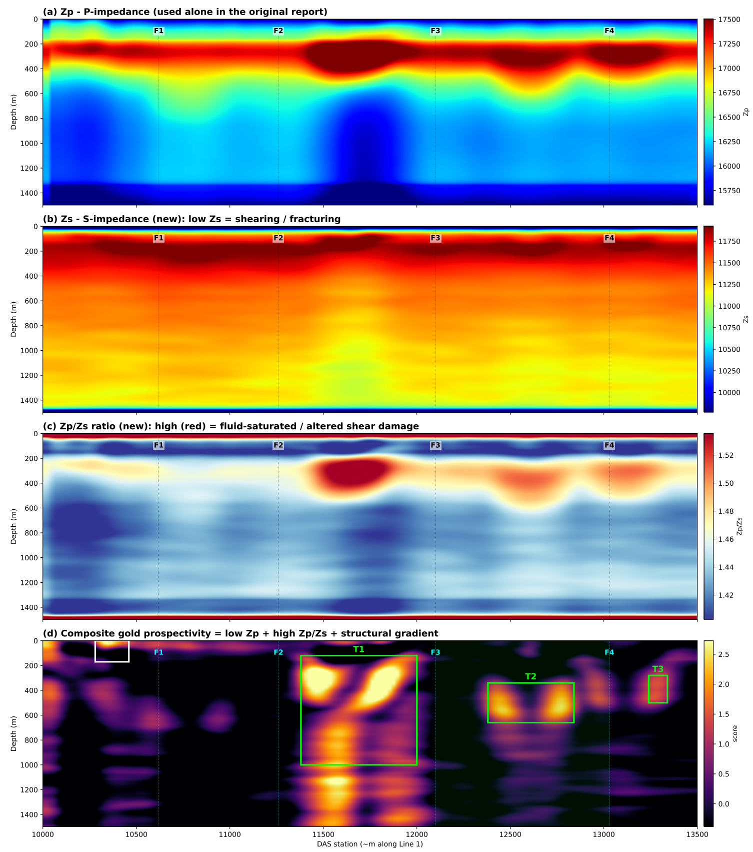

Figure 1: Dual-impedance re-analysis of Line 1: (a) Zp; (b) Zs (new); (c) Zp/Zs ratio (new); (d) composite gold prospectivity with targets T1–T3. Shear zones F1–F4 after the original report. Impedances are relative (uncalibrated).

6. Georeferenced structural integration

The GM-49463 1:10,000 compilation carries a latitude/longitude graticule (top edge 77°30′00″–77°22′30″ W at 48°22′30″ N). A pixel-to-UTM transform built from the neat-line corners (reconstructing the NE corner to the metre) georeferences the map; the principal Manneville lineament, digitized and converted to UTM, yields a strike of N61°W (fit $N = -0.559E + 5{,}535{,}051$). The trace runs ∼1.2 km south of Line 1's southwest end (3–4.8 km south of the centre/northeast), so Line 1 lies wholly on the northern (hanging-wall) flank — the side on which historical mapping places the auriferous branch shears (Bischoff, 1965). The drilled gold centres (Pershing-Manitou, Zone A, Uniacke) lie northwest of and along strike toward the property (Fig. 2). The NW (N45°W) gold-branch trend projects onto the northeast-central part of Line 1 (DAS ≈10,500–12,000), overlapping the northeast edge of T1. Absolute georeferencing accuracy is $\sim \pm$100–150 m (NAD27 corner-only graticule, scan distortion, fault curvature).

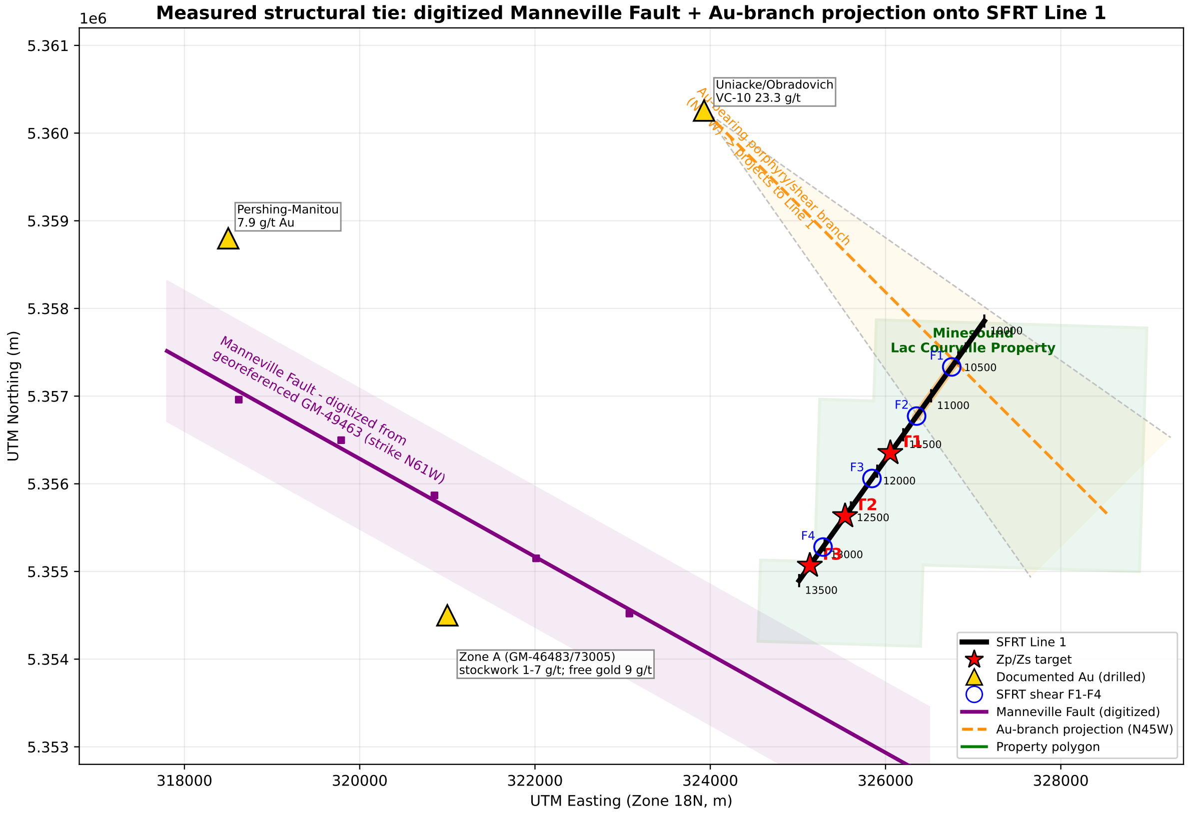

Figure 2: Measured structural tie (UTM 18N): digitized Manneville Fault (purple, N61°W) from georeferenced GM-49463; documented gold occurrences; the N45°W gold-branch projection (orange) onto Line 1; property polygon; Line 1 with shear zones F1–F4 and targets T1–T3.

7. Pre-drill predictions and falsification criteria

We register the following predictions before drilling (Table 2). For each hole the predicted target is a discrete deformation zone (foliated, quartz-veined) carrying sericite-carbonate $\pm$ silicification and pyrite $\pm$ chalcopyrite, at the depth window given by the Zp/Zs target. Two criteria are defined and will be assessed independently:

(T1) Structural-hit criterion. The hole intersects a shear/altered/veined zone within $\pm$50 m (true vertical) of the predicted depth window.

(T2) Mineralization-hit criterion. Gold $\geq$0.3 g/t over $\geq$1 m occurs within, or immediately adjacent to ($\leq$10 m of), that structural zone.

Program-level outcomes (pre-registered). (a) If $\geq$2 of 3 holes meet the structural-hit criterion, SFRT is judged effective for *structural* targeting on this line. (b) If $\geq$1 of 3 holes meets the mineralization-hit criterion, SFRT is judged to show *direct mineralization vectoring*. (c) If 0 of 3 holes meets the structural-hit criterion, the test is recorded as a failure of SFRT structural targeting at Lac Courville. A single barren but structurally-correct intersection does not, by itself, refute the method, because target T2 addresses structure rather than assured fertility.

Table 2: Pre-registered predictions and pass/fail criteria (filed before drilling).

Hole (target)	Predicted depth window	Structural-hit (T1 criterion)	Mineralization-hit (T2 criterion)
DDH-LC-01 (T1)	250–550 m	shear/altered/veined zone within $\pm$50 m	Au $\geq$0.3 g/t / $\geq$1 m in or $\leq$10 m from the zone
DDH-LC-02 (T2)	340–660 m (tested to 544 m)	shear/altered/veined zone within $\pm$50 m	Au $\geq$0.3 g/t / $\geq$1 m in or $\leq$10 m from the zone
DDH-LC-03 (T3)	280–500 m	shear/altered/veined zone within $\pm$50 m	Au $\geq$0.3 g/t / $\geq$1 m in or $\leq$10 m from the zone

Table 3: Planned drill holes (UTM Zone 18N; azimuth grid; collar elevations to be set by RTK).

Hole	Target	Collar E	Collar N	Azimuth	Dip	Length
DDH-LC-01	T1	326,217	5,356,515	225°	$-60°$	600 m
DDH-LC-02	T2	325,703	5,355,797	225°	$-65°$	600 m
DDH-LC-03	T3	325,307	5,355,239	225°	$-58°$	600 m

8. Drilling program and test protocol

Three NQ diamond-drill holes of 600 m each are planned (Table 3; Fig. 3). All are oriented at azimuth 225° (grid), perpendicular to the N45°W gold-branch strike, with collars northeast of and up-dip from each target so the hole drills southwest-and-down, cutting the steeply NE-dipping shears at a high angle. Dips ($-58°$ to $-65°$) are chosen so a 600 m hole brackets each target. We note explicitly that a 600 m hole cannot reach the base of T2 (660 m); DDH-LC-02 therefore tests the upper-central part of T2 (to $\sim$544 m vertical, which contains the anomaly centroid), and an extension to $\sim$730 m would be required to test its base. Grid convergence in the area is $\sim -1.7°$ (true azimuth $\approx$223.3°); magnetic declination ($\sim -14°$, 2026) must be applied for compass orientation. Collar elevations will be fixed by RTK survey.

Quality assurance and third-party verification. Core will be logged and signed off by an independent Qualified Person (Québec-registered, NI 43-101 competent). Samples will be assayed at an ISO/IEC 17025-accredited laboratory (Activation Laboratories, Actlabs) by fire assay with AA finish (gravimetric finish for high grades), with certified standards, blanks and duplicates inserted for QA/QC. Collars and hole traces will be surveyed by RTK-GPS. Core will be photographed and archived. These measures, together with the time-stamped predictions, are intended to allow external parties to verify the test independently of the authors.

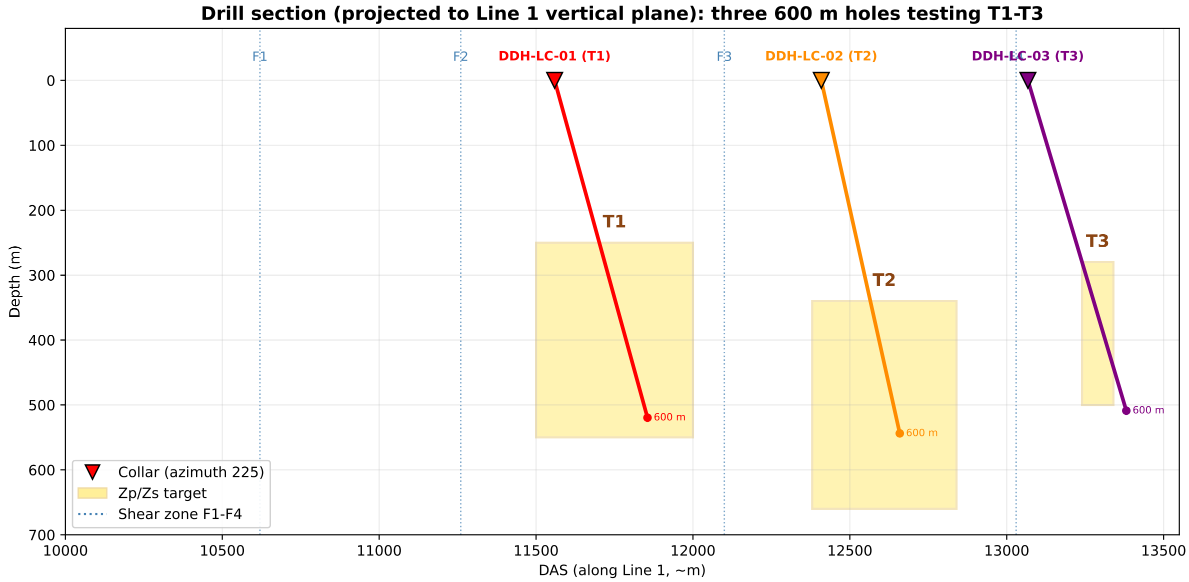

Figure 3: Drill section projected onto the Line 1 vertical plane: three 600 m holes (collars at azimuth 225°) testing targets T1–T3. Gold boxes are the Zp/Zs targets; dotted lines are shear zones F1–F4.

9. Discussion

The test is deliberately conservative in what a positive result can claim. Because Line 1's impedances are relative, the Zp/Zs anomalies indicate favourable *rock state* (alteration, fracturing, fluid), not assayed grade; a structural hit is therefore the primary, most defensible outcome, and mineralization is the more demanding secondary outcome. The single-line geometry constrains targets in depth and along-line position but not across strike, so a near-miss in the cross-line direction cannot be excluded for any individual hole; this is one reason the program-level criteria require a majority of holes rather than relying on one. Absolute georeferencing of the historical corridor ($\sim \pm$100–150 m) does not affect the on-line targets, which derive directly from the GPS-located Line 1, but it does limit how precisely the external gold occurrences can be tied to specific drill intersections.

The principal threat to validity we cannot fully remove is the authors' dual role as property holder and method provider. We mitigate it structurally: the predictions are time-stamped and public before drilling; the criteria are quantitative and fixed; and adjudication relies on an independent QP, an accredited laboratory, and RTK survey rather than on the authors' judgement. We also commit to reporting Part II regardless of outcome, including a null result.

10. Conclusions

We have re-analysed a single SFRT profile over the Lac Courville Property using the density-independent Zp/Zs ratio, integrated it with a georeferenced compilation of the district's documented gold endowment and the digitized Manneville Fault, and used the result to define three ranked targets and a three-hole, 600 m drill program. Crucially, we register the targets, drill-hole geometry, and quantitative pass/fail criteria *before* drilling. Part II will report the drilling and adjudicate the pre-registered criteria without modification, providing a falsifiable, externally witnessed assessment of whether SFRT adds value for orogenic-gold targeting — an assessment whose credibility does not depend on the outcome.

Data and code availability

All raw and processed data (impedance volumes, property polygon, targets, drill plan, predictions, reference ledger), the processing and figure code, the figures, and the LaTeX source of this manuscript are archived in the project repository https://github.com/Ruqing1963/lac-courville-sfrt-gold-test and released with a versioned DOI on Zenodo at the time of submission, establishing the time-stamp of the pre-registered predictions.

Acknowledgments

We thank the Québec Ministère des Ressources naturelles et des Forêts (SIGÉOM/EXAMINE) for open access to the historical assessment-report archive on which the district compilation depends, and the original authors of those reports, whose work is cited individually in the repository ledger.

Author contributions

A. Xue developed the SFRT method and acquired/processed the Line 1 data. R. Chen compiled the historical record, performed the georeferencing, structural integration and drill design, and drafted the manuscript. Both authors approved the pre-registered protocol.

Competing interests

R. Chen is affiliated with Minesound Ltd. (property holder); A. Xue is affiliated with the SFRT method provider. No other competing interests are declared. The pre-registration and third-party verification described in §8 are intended to mitigate these interests.

References

A.-J. Beauregard. Rapport de compilation, propriété courville, projet pershing-manitou (#410). Technical report, Géologica for Placer Dome Inc.; assessment report GM-49463, MRNF Québec, 1990.

C.T. Bischoff. Engineer's report and geology, big town copper mines ltd. Technical report, assessment report GM-16070, MRNF Québec, 1965.

B. Dubé and P. Gosselin. Greenstone-hosted quartz-carbonate vein deposits. In W.D. Goodfellow, editor, *Mineral Deposits of Canada: a synthesis of major deposit-types, district metallogeny, the evolution of geological provinces, and exploration methods*, number 5 in Geological Association of Canada, Mineral Deposits Division, Special Publication, pages 49–73. 2007.

Globex Mining Enterprises Inc. Courville property description. https://globexmining.com/property/courville/, n.d. accessed 2026.

P.J. Hawley. Report on the prospecting program, uniacke prospect, courville township. Technical report, for T. Obradovich; assessment report GM-53064, MRNF Québec, 1994.

LaFleur Minerals Inc. News release: acquisition of mineral claims, mckenzie east gold project, val-d'or district. company news release, 2026. regional context for the Pascalis-Tiblemont batholith and Manneville Deformation Corridor.

G. Laverdière. NI 43-101 technical report on the courville gold project, abitibi, québec. Technical report, Opus One Gold Corp.; filed as assessment report GM-73005, MRNF Québec, 2021.

J. Patel. Report on diamond drilling, courville township property (zone a). Technical report, Cache d'Or Resources Inc.; assessment report GM-46483, MRNF Québec, 1988.

Aimin Xue. Method for exploring passive source seismic frequency resonance. U.S. Patent 11,650,342 B2, 2023. Priority Chinese Patent Application CN201811587566.3, filed 25 Dec. 2018.

Aimin Xue et al. Seismic frequency resonance prospecting – a new geophysical exploration technique. American Geophysical Union, Fall Meeting 2020, abstract T033-0013, 2020. Bibcode 2020AGUFMT033.0013X.